



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

01-08-2021

JEE(Main + Advanced) : Enthusiast Course (PHASE : TIAS, S,I(A),I, LIVE-I & I(A))
ANSWER KEY**PAPER-1****PART-1 : PHYSICS**

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	A,B,C,D	A,C,D	A,B,C,D	A,B,C	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	C	A	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.66 to 1.67	15.00	1.00	1.20	0.00	5.25	0.86 to 0.87	369.30

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	A,C	A,B,C	B,D	A,B,C,D	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	B	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	- 200.00	615.00	2.40	29.00	223.40	4.50	9.00	4.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	B,C,D	A,B,C	A,D	A,B,C,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	C	A	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	1.12 or 1.13	3.00	3.00	37.33	14.00	3.00	24.50

HINT – SHEET**PART-1 : PHYSICS****SECTION-I (i)****Ans (A,B,C)**

$$1. VP^n = \text{constant} \Rightarrow P^n dV + nP^{n-1} V dP = 0$$

$$\frac{-P}{nV} = \frac{dP}{dV}$$

$$\beta = -\frac{V dP}{dV} = \frac{P}{n}$$

$$\text{Volume exp. coefficient} = \frac{1}{V} \frac{dV}{dT}$$

$$VP^n = \text{constant}$$

$$\frac{P^n V^n}{V^{n-1}} = \text{constant} \Rightarrow T^n V^{1-n} = \text{constant}$$

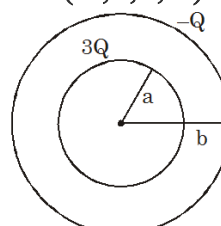
$$nT^{n-1} V^{1-n} dT + (1-n) T^n V^{-n} dV = 0$$

$$\frac{nVdT}{dT} + (1-n) \frac{TdV}{dV} = 0$$

$$\frac{dT}{dT} = \frac{(n-1)T}{nV}$$

So volume exp. coefficient =

$$\frac{1}{V} \left(\frac{nV}{(n-1)T} \right) = \frac{n}{(n-1)T}$$

2. Ans (A,B,C,D)When $a < R < b$
Electric field due to $3Q$ only

$$\text{So } E = \frac{k(3Q)}{R^2} = \frac{3Q}{4\pi\epsilon_0 R^2}$$

and

$$V = \frac{k(3Q)}{R} + \frac{k(-Q)}{b}$$

$$V = \frac{Q}{4\pi\epsilon_0} \left(\frac{3}{R} - \frac{1}{b} \right)$$

When $R > b$

$$E = \frac{k(3Q - Q)}{R^2} = \frac{2Q}{4\pi\epsilon_0 R^2}$$

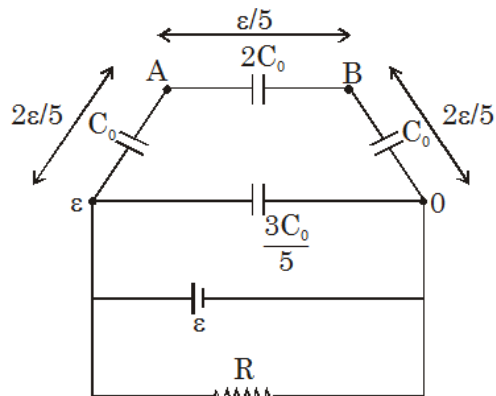
$$\text{and } V = \frac{k(3Q - Q)}{R^2} = \frac{2Q}{4\pi\epsilon_0 R}$$

3. Ans (A,C,D)

Terminals of resistance are directly connected with the terminals of ideal battery so potential difference across resistance remains constant.

So the current in it.

Simplified circuit.



$$Q_{C_3} = \left(\frac{3C_0}{5} \right) \varepsilon = \frac{3C_0 \varepsilon}{5}$$

$$\Delta V_{AB} = \frac{\varepsilon}{5}$$

4. Ans (A,B,C,D)

(A) As net force on satellite is towards earth so the acceleration.

(B) No torque on satellite about centre of earth so.

(C) KE at a point in orbit is constant but different at different point, when satellite reaches to a point again then its KE becomes equal when it was at last time when it was at same point.

(D) ME $\rightarrow$ is conserved as no non conservative force is acting.

5. Ans (A,B,C)

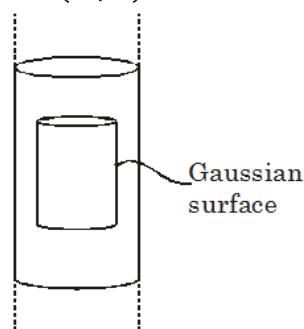
$$R = \frac{\rho \ell}{A}$$

$$i = \frac{\varepsilon}{R} = \frac{\varepsilon A}{\rho \ell}$$

$$E = \frac{\varepsilon}{\ell}$$

$$i = neAV_d \Rightarrow V_d = \frac{i}{neA} = \frac{\varepsilon}{\rho \ell n e}$$

6. Ans (A,B)



From gauss law (inside)

$$E (2\pi r \ell) = \frac{(\pi r^2 \ell) \rho}{\varepsilon_0}$$

$$E = \frac{r \rho}{2\varepsilon_0} \Rightarrow E \propto r$$

Outside

$$E (2\pi r \ell) = \frac{\pi R^2 \ell \rho}{\varepsilon_0}$$

$$E = \frac{R^2 \rho}{2r \varepsilon_0} \quad E \propto r^{-1}$$

SECTION-I (ii)

7. Ans (C)

To solve this problem we can apply principle of superposition. Let us assume current I enters at A and spreads out in the form of hemispherical region. So current density at a distance r from

$$A = \frac{I}{2\pi r^2}$$

Now using Ohm's law, $E = J\rho$

$$E = \frac{I\rho}{2\pi r^2}$$

$$\text{or } \frac{-dV}{dr} = \frac{I\rho}{2\pi r^2}$$

$$\int_{V_D}^{V_C} dV = \frac{-I\rho}{2\pi} \int_a^{a\sqrt{2}} \frac{dr}{r^2}$$

$$V_C - V_D = \frac{-I\rho}{2\pi} \left[\frac{-1}{r} \right]_a^{a\sqrt{2}}$$

$$V_C - V_D = \frac{I\rho}{2\pi} \left[\frac{1}{a\sqrt{2}} - \frac{1}{a} \right]$$

$$V_D - V_C = \frac{I\rho}{2\pi a} \left[\frac{\sqrt{2}-1}{\sqrt{2}} \right]$$

Net potential difference between D and C

$$= 2 \times \frac{I\rho}{2\pi a} \left[\frac{\sqrt{2}-1}{\sqrt{2}} \right]$$

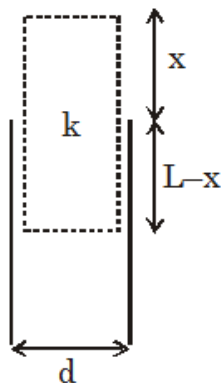
8. **Ans (C)**

Emf

$$E = V + Ir$$

$$E = \frac{I\rho}{\pi a} \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right) + Ir$$

9. **Ans (A)**



Let plates are dimension of $L \times L$ as battery is disconnected so charge on plates remains constant.

Now

$$C = \frac{(x\ell)\epsilon_0}{d} + \frac{k(\ell-x)\ell\epsilon_0}{d}$$

$$C = \frac{\ell\epsilon_0}{d} (x + k\ell - kx)$$

for charge conservation

$$CV = C_0 V_0$$

$$V = \left(\frac{k\ell^2\epsilon_0}{d} \right) \times \left(\frac{d}{\ell\epsilon_0(k\ell+x)(1-k)} \right) V_0$$

$$V = \frac{k\ell V_0}{k\ell+x(1-k)} \quad \{k > 1\}$$

So B is correct.

10. **Ans (D)**

As battery remains connected so potential difference across capacitor remains constant.

SECTION-II

1. **Ans (1.66 to 1.67)**

$$\frac{E_1 + E_2}{E_1 - E_2} = \frac{8}{2} \Rightarrow 2E_1 + 2E_2 = 8E_1 - 8E_2$$

$$6E_1 = 10E_2 \Rightarrow \frac{E_1}{E_2} = \frac{5}{3} = 1.6667$$

2. **Ans (15.00)**

Process

$i \rightarrow 2$ and process

$3 \rightarrow 4$ are isochoric processes.

$$W_{12} = 0, W_{34} = 0, W_{23} = nR(T_3 - T_2) = 3R(1600 - 400)$$

$$W_{41} = nR(T_1 - T_4) = 3R(200 - 800) = -1800R$$

$$W = (3600 - 1800)R = 1800R = 15 \text{ kJ}$$

3. **Ans (1.00)**

When both identical spheres are touched potential of both becomes equal as both are identical in shape so charge also.

Charge on A

$$\text{After 1st touch } \frac{Q_0 + 0}{2} = \frac{Q_0}{2}$$

$$\text{After 2nd touch } \frac{\frac{Q_0}{2} + Q_0}{2} = \frac{3Q_0}{4}$$

$$\text{After 3rd touch } \frac{\frac{3Q_0}{4} + Q_0}{2} = \frac{7Q_0}{8}$$

$$\text{After 4th touch } \frac{\frac{7Q_0}{8} + Q_0}{2} = \frac{15Q_0}{16}$$

$$\text{After nth touch } \frac{(2^n - 1)Q_0}{2^n}$$

$$\text{So } x = 2 = 4 \text{ or } \frac{x}{4} = 1$$

4. **Ans (1.20)**

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{-1} - \frac{1.5}{u} = \frac{1 - 1.5}{-2}$$

$$-1 - \frac{1}{u} = \frac{3}{2u}$$

$$-\frac{5}{4} = \frac{3}{2u}$$

$$u = -\frac{12}{10} = -1.2 \text{ cm}$$

5. **Ans (0.00)**

Potential drop across ammeter is zero as it is an ideal ammeter

So $V = 0$

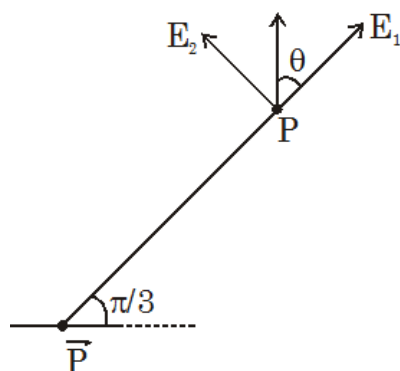
$$\text{and } \frac{V}{i} = 0$$

6. Ans (5.25)

If P.D. across each bulb is 1.5 V then P.D. across resistance must be $12 - 1.5 = 10.5$ and current in it is 2A as in each bulb current is 0.5 A.

$$\text{So } R = \frac{10.5}{2} = 5.25\Omega$$

7. Ans (0.86 to 0.87)



$$E_1 = \frac{\frac{2kP}{2}}{r^3}$$

$$E_2 = \frac{\frac{k\sqrt{3}P}{2}}{r^3}$$

$$\tan \theta = \frac{E_2}{E_1} = \frac{\sqrt{3}}{2}$$

$$\theta = \tan^{-1} \left(\frac{1.732}{2} \right)$$

$$\theta = \tan^{-1} (0.866)$$

$$\text{From x-axis angle} = \frac{\pi}{3} + \tan^{-1} (0.87)$$

8. Ans (369.30)

According to the first law of thermodynamics,

$$\Delta Q = \Delta U + \Delta W$$

For the vessel, for which internal energy

remains constant, $\Delta U = 0$

$$\therefore \Delta Q_1 = \Delta W = \mu RT \log_e \left(\frac{V_F}{V_I} \right)$$

$$= 1 \times R \times 300 \times \log_e 2$$

$$= 207.9 R$$

For the vessel, for which volume is kept constant

$$\Delta Q_2 = \Delta U \quad (\because \Delta W = 0)$$

$$= \mu C_V \Delta T$$

$$= 1 \times \frac{3}{2} R \times \Delta T$$

Since $\Delta Q_1 = \Delta Q_2$ (given)

$$207.9R = \left(\frac{3}{2} \right) R \Delta T$$

$$\therefore \Delta T = 138.6$$

$$\therefore T_f - T_i = \Delta T = 138.6$$

$$\therefore T_f = 300 + 138.6 = 438.6 \text{ K}$$

When mixing of gas takes place,

$U_1 + U_2 = U$ (by law of conservation of energy)

$$\therefore \mu_1 (C_V)_1 T_1 + \mu_2 (C_V)_2 T_2 = \mu C_V T$$

Now $\mu = \mu_1 + \mu_2$ (by law of conservation of mass)

Here $\mu_1 = \mu_2 = 1$ and $(C_V)_1 = (C_V)_2 = C_V$

$$\therefore 1 \times 300 + 1 \times 438.6 = 2T$$

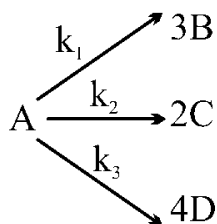
$$\therefore T = 369.3 \text{ K}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A, C, D)

$$t = 0 \quad 1M$$



$$t = \infty \quad 0$$

$$(B) [B_t] : [C_t] \quad \therefore k_1 \times 3 : k_2 \times 2$$

$$\therefore 1 \times 3 : 2 \times 2$$

$$\therefore 3 : 4$$

$$\therefore [B_t] < [C_t]$$

$$(C) [D_t] = 4 \times \frac{3}{6} \times \frac{1}{2} = 1M$$

SECTION-I (ii)

7. Ans (D)

$$(A) P_1 V_1^3 = P_2 V_2^3 \Rightarrow 1 V_1^3 = 27 \times V_2^3 = V_1 = 3 V_2$$

$$\Rightarrow V_2 = \frac{24}{3} = 8 \text{ lit.}$$

$$(B) \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \Rightarrow T_2 = 2700 \text{ K}$$

$$(C) W = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = \frac{27 \times 8 - 1 \times 24}{3 - 1} = 96 \text{ lit atm}$$

8. Ans (B)

$$C_m = C_v + \frac{R}{1 - x} \Rightarrow \frac{3R}{2} + \frac{R}{1 - (-1)} \Rightarrow 2R$$

$$T_V^{-2} = k$$

$$T_V^{-2} = T_V^{(n-1)} \Rightarrow x - 1 = -2 \Rightarrow x = -1$$

SECTION-II

1. Ans (- 200.00)

$$2(-16) = 2(-46) - 300 (\Delta_r S^\circ)$$

$$\Delta_r S^\circ = 200 \text{ J/K}$$

2. Ans (615.00)

$$3$$

3. Ans (2.40)

$$E = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1168 \times 10^{-10}} = 1.69 \times 10^{-18} \text{ J}$$

$$= 10.6 \text{ eV}$$

$$KE = E_{\text{photon}} - W = 10.6 - 5.6 = 5 \text{ eV}$$

4. Ans (29.00)

Malachite $[\text{CuCO}_3 \cdot \text{Cu(OH)}_2]$ is ore of copper (atomic number $Z = 29$).

6. Ans (4.50)

Sum of $n + \ell + m + s$ for $4s^1$ is $4 + 0 + 0 + \frac{1}{2} = 4.5$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A, C)

Note all 3 limits are of $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form, so by

L'Hôpital's rule, we have

$$\frac{1}{x^2 + e^{2x}} \cdot (2x + 2e^{2x})$$

$$L = \lim_{x \rightarrow \lambda} \frac{\frac{1}{x^2 + e^{2x}} \cdot (2x + 2e^{2x})}{\frac{1}{x^6 + e^x} (6x^5 + e^x)}, \{\lambda = 0, \pm\infty\}$$

$$p = \lim_{x \rightarrow 0} \frac{\frac{1}{0+1} (0+2)}{\frac{1}{0+1} (0+1)} = 2$$

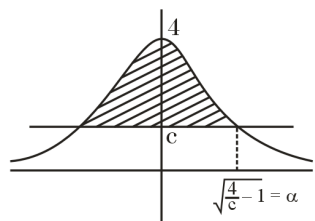
$$q = \lim_{x \rightarrow \infty} \frac{2 \left(\frac{x}{e^{2x}} \right) + 2 \left(\frac{x^6}{e^x} \right) + 1}{\frac{x^2}{e^{2x}} + 1} \cdot \frac{\left(\frac{x^6}{e^x} \right) + 1}{6 \left(\frac{x^5}{e^x} \right) + 1} = 2 \left(\text{as } \lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0 \right)$$

$$r = \lim_{x \rightarrow -\infty} \frac{2 + 2 \left(\frac{e^{2x}}{x} \right)}{x \left\{ 1 + \left(\frac{e^{2x}}{x^2} \right) \right\}} \cdot \frac{x \left\{ 1 + \frac{e^x}{x^6} \right\}}{\left\{ 6 + \frac{e^x}{x^5} \right\}}$$

$$= \frac{2}{6} = \frac{1}{3} \left(\text{as } \lim_{x \rightarrow -\infty} \frac{e^x}{x^n} = 0 \forall n > 0 \right)$$

$$\text{Hence } p = q = 2 \text{ \& } r = \frac{1}{3}$$

2. Ans (B, C, D)



$$2 \int_0^{\alpha} \left(\frac{4}{1+x^2} - c \right) dx = \int_0^{\infty} \frac{4}{1+x^2} dx$$

$$\Rightarrow 8 \tan^{-1} \alpha - 2c\alpha = 2\pi$$

$$\Rightarrow 4 \tan^{-1} \left(\sqrt{\frac{4}{c} - 1} \right) - \sqrt{4c - c^2} = \pi$$

$$\text{Now } 4 \tan^{-1} \left(\sqrt{\frac{4}{c} - 1} \right) > \pi$$

$$\Rightarrow \frac{4}{c} - 1 > 1 \Rightarrow c < 2$$

Note as c decreases LHS increases at $c = 1$

$$\text{LHS} = 4 \tan^{-1} \sqrt{3} - \sqrt{3} = \frac{4\pi}{3} - \sqrt{3} < \pi$$

$$\Rightarrow \frac{\pi}{3} < \sqrt{3} \text{ (True)}$$

Hence for LHS = π , $c < 1$

So (B), (C)

3. Ans (A, B, C)

$$y e^{-x} = A + B e^x + C e^{2x}$$

$$\Rightarrow y' e^{-x} - y e^{-x} = B e^x + 2C e^{2x}$$

$$\Rightarrow y' e^{-2x} - y e^{-2x} = B + 2C e^x$$

$$\Rightarrow y'' e^{-2x} - 3y' e^{-2x} + 2y e^{-2x} = 2C e^x$$

$$\Rightarrow y'' e^{-3x} - 3y' e^{-3x} + 2y e^{-3x} = 2C$$

$$\Rightarrow y''' e^{-3x} - 6y'' e^{-3x} + 11y' e^{-3x} - 6y e^{-3x} = 0$$

$$\Rightarrow \frac{d^3 y}{dx^3} - 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} - 6y = 0$$

4. Ans (A, D)

$$\int \frac{x}{\cos x} \cdot \frac{x \cos x}{(x \sin x + \cos x)^2} dx$$

$$= \frac{x}{\cos x} \left(-\frac{1}{x \sin x + \cos x} \right) - \int \left(\frac{\cos x + x \sin x}{\cos^2 x} \right) \left(\frac{-1}{x \sin x + \cos x} \right) dx$$

$$= -\frac{x}{\cos x (x \sin x + \cos x)} + \frac{\sin x}{\cos x} + C$$

$$= \frac{-x + x \sin^2 x + \sin x \cos x}{\cos x (x \sin x + \cos x)} + C$$

$$= \frac{\sin x \cos x - x \cos^2 x}{\cos x (x \sin x + \cos x)} + C$$

$$= \frac{\tan x - x}{1 + x \tan x} + C$$

$$= \tan(x - \tan^{-1} x) + C$$

5. Ans (A, B, C, D)

Since no information is provided about $f(a)$

hence nothing definite can be concluded about

the behaviour of this function at $x = a$,

we cannot even comment whether $f(x)$ is

defined at $x = a$ or not.

Hence A, B, D are definitely incorrect.

Now $\lim_{t \rightarrow 0} \{f(a+t) - f(a-t)\} = 0$, does not

ensure that $\lim_{x \rightarrow a} f(x)$ exists (for example

consider $f(x) = \ln |x - a|$ at $x = a$, note $\lim_{t \rightarrow 0} (\ln |a$

$+ t - a| - \ln |a - t - a|) = 0$ but $\lim_{x \rightarrow a} f(x)$ does not

exists).

Hence all the options (A), (B), (C), (D) are

incorrect.

6. Ans (A, C)

$$P = \lim_{x \rightarrow 0^+} \frac{\sin |x| \cdot 2x}{\sec^2 x - 1} \text{ (L'Hôpital's Rule)}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sin x \cdot 2x}{\frac{\tan^2 x}{x^2} \cdot x \cdot x} = 2$$

$$Q = \lim_{x \rightarrow 0^-} \frac{\sin |x| \cdot 2x}{\sec^2 x - 1} \text{ (L'Hôpital's Rule)}$$

$$= \lim_{x \rightarrow 0^-} \frac{\sin x \cdot 2x}{\frac{\tan^2 x}{x^2} \cdot x \cdot x} = -2$$

Since $P \neq Q$, L does not exist.

SECTION-I (ii)

7. Ans (C)

$$f(x+y) \cdot f(x-y) = f^2(x) \quad \dots(1)$$

Diff. w.r.t. x (taking y as constant)

$$f'(x+y) \cdot f(x-y) + f(x+y) \cdot f'(x-y) = 2f(x)$$

$$\cdot f'(x)$$

Substitute $x = 0$

$$f'(y) \cdot f(-y) + f(y) \cdot f'(-y) = 2 \ln 2$$

$$\left\{ \text{Now substitute } x = 0 \text{ in (1) to get } f(-y) = \frac{1}{f(y)} \right\}$$

$$\text{So } \frac{f'(y)}{f(y)} + \frac{f'(y)}{f(y)} = 2 \ln 2 \Rightarrow \frac{f'(y)}{f(y)} = \ln 2$$

$$\Rightarrow f(y) = 2^y \text{ (as } f(0) = 1)$$

$$\Rightarrow f(x) = 2^x \text{ \& } f^{-1}(x) = \log_2 x$$

Hence one-one onto

$$\int_0^2 \frac{\ln x}{\ln 2} dx = \frac{x \ln x - x}{\ln 2} \Big|_0^2$$

$$= \frac{2 \ln 2 - 2}{\ln 2} = 2 - 2 \log_2 e = \log_2 \left(\frac{4}{e^2} \right)$$

8. Ans (C)

$$f(x+y) \cdot f(x-y) = f^2(x) \quad \dots(1)$$

Diff. w.r.t. x (taking y as constant)

$$f'(x+y) \cdot f(x-y) + f(x+y) \cdot f'(x-y) = 2f(x)$$

$$\cdot f'(x)$$

Substitute $x = 0$

$$f'(y) \cdot f(-y) + f(y) \cdot f'(-y) = 2 \ln 2$$

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$$\text{So } \frac{f'(y)}{f(y)} + \frac{f'(y)}{f(y)} = 2 \ln 2 \Rightarrow \frac{f'(y)}{f(y)} = \ln 2$$

$$\Rightarrow f(y) = 2^y \text{ (as } f(0) = 1)$$

$$\Rightarrow f(x) = 2^x \text{ \& } f^{-1}(x) = \log_2 x$$

Hence one-one onto

$$\int_0^2 \frac{\ln x}{\ln 2} dx = \frac{x \ln x - x}{\ln 2} \Big|_0^2$$

$$= \frac{2 \ln 2}{\ln 2} = 2 - 2 \log_2 e = \log_2 \left(\frac{4}{e^2} \right)$$

9. Ans (A)

$$f(x) = \begin{cases} \pi - 2 + \pi - 3 + (2k+1)\pi - x \\ \pi - 2 + \pi - 3 + x - 2k\pi \end{cases}$$

$$\text{If } x \in \left[\frac{3\pi}{2}, \frac{5\pi}{2} \right]$$

$$f(x) = 2\pi - 5 + x - 2\pi = x - 5$$

so for $x = 5, 6, 7$ $f(x)$ is an integer

$$\text{Hence } \sum_{r=1}^3 f(x_r) = \sum_{r=1}^3 x_r - 15 = 18 - 15 = 3$$

$$f'(x) = \begin{cases} -1 & x \in (2k\pi + \frac{\pi}{2}, 2k\pi + \frac{3\pi}{2}) \\ 1 & x \in (2k\pi + \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}) \end{cases} \quad k \in \mathbb{Z}$$

$$\text{Hence } f'(r) = 1 \text{ for } r = 1, 5, 6, 7 \text{ \&}$$

$$f'(r) = -1 \text{ for } r = 2, 3, 4, 8, 9, 10$$

$$\text{So } \sum_{r=1}^{10} f'(r) = -2$$

10. Ans (D)

$$f(x) = \begin{cases} \pi - 2 + \pi - 3 + (2k+1)\pi - x \\ \pi - 2 + \pi - 3 + x - 2k\pi \end{cases}$$

$$\text{If } x \in \left[\frac{3\pi}{2}, \frac{5\pi}{2} \right]$$

$$f(x) = 2\pi - 5 + x - 2\pi = x - 5$$

so for $x = 5, 6, 7$ $f(x)$ is an integer

$$\text{Hence } \sum_{r=1}^3 f(x_r) = \sum_{r=1}^3 x_r - 15 = 18 - 15 = 3$$

$$f'(x) = \begin{cases} -1 & x \in (2k\pi + \frac{\pi}{2}, 2k\pi + \frac{3\pi}{2}) \\ 1 & x \in (2k\pi + \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}) \end{cases} \quad k \in \mathbb{Z}$$

$$\text{Hence } f'(r) = 1 \text{ for } r = 1, 5, 6, 7 \text{ \&}$$

$$f'(r) = -1 \text{ for } r = 2, 3, 4, 8, 9, 10$$

$$\text{So } \sum_{r=1}^{10} f'(r) = -2$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1.00)**

$$y = e^{-x+n}, x \in [n, n+1) \quad n \in \mathbb{Z}$$

Equation of tangent at (x_1, e^{-x_1+n}) is

$$y - e^{-x_1+n} = -e^{-x_1+n} (x - x_1)$$

Passes through $(-\frac{1}{2}, 0)$, hence

$$0 - e^{-x_1+n} = -e^{-x_1+n} \left(-\frac{1}{2} - x_1 \right)$$

$$\Rightarrow x_1 = -\frac{1}{2}$$

Hence only one tangent passes through

$$\left(-\frac{1}{2}, 0 \right)$$

2. **Ans (1.12 or 1.13)**

$$A = \sum_{r=1}^9 \cot^{-1} \left(\frac{r-5}{r(r-10)+50} \right)$$

$$A = \sum_{r=1}^9 \cot^{-1} \left(\frac{10-r-5}{(10-r)(10-r-10)+50} \right) \left\{ \sum_{r=m}^n f(r) = \sum_{r=m}^n f(m+n-r) \right\}$$

$$\Rightarrow 2A = \sum_{r=1}^9 \left\{ \cot^{-1} \left(\frac{r-5}{r^2-10r+50} \right) + \cot^{-1} \left(\frac{5-r}{r^2-10r+50} \right) \right\}$$

$$\text{Hence } A = \frac{9\pi}{2}$$

$$\text{Now } B = \int_1^9 \cot^{-1} \left(\frac{x-5}{x^2-10x+50} \right) dx$$

$$\Rightarrow B = \int_1^9 \cot^{-1} \left(\frac{5-x}{x^2-10x+50} \right) dx$$

$$\Rightarrow 2B = \int_1^9 \pi dx \Rightarrow B = 4\pi$$

$$\text{So } \frac{A}{B} = \frac{9}{8} = 1.125$$

3. **Ans (3.00)**

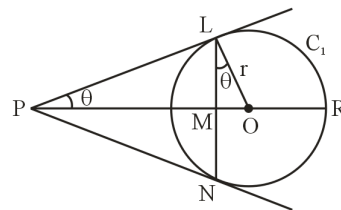
$$\text{Consider } H(x) = \frac{f'(x)}{f(x)} - x$$

$H(1), H(2), H(3), H(4) = 0$ by Rolle's Theorem

$$H'(x) = \frac{f(x) \cdot f''(x) - (f'(x))^2}{f^2(x)} - 1 = 0$$

has atleast 3 roots in $(1, 4)$

4. **Ans (3.00)**



$$PQ = 6$$

$$OM = r \cos \theta$$

$$QM = 2 = r(1 - \sin \theta)$$

$$PQ = 6 = r(\operatorname{cosec} \theta - 1)$$

$$\text{Hence } \frac{1}{3} = \sin \theta$$

$$PQ = 6 = r(3 - 1)$$

$$\Rightarrow r = 3$$

ALITER :

PM is Harmonic Mean of PQ and PR

$$\text{So } 8 = \frac{2.6(6+2r)}{6+6+2r} = \frac{24(r+3)}{2(r+6)}$$

$$\Rightarrow 2r + 12 = 3r + 9 \Rightarrow r = 3$$

5. **Ans (37.33)**

$$\int_1^2 \frac{6x^5 + 5x^4 + 4x^3}{x^6 + x^5 + x^4} dx$$

(Multiplying N^r & D^r of integrand by x^3)

$$= \ln(x^6 + x^5 + x^4) \Big|_1^2 = \ln\left(\frac{112}{3}\right)$$

$$\Rightarrow k = 37.33$$

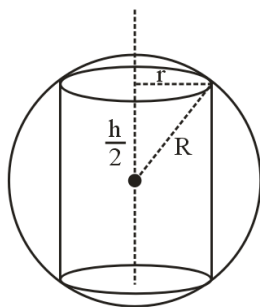
6. **Ans (14.00)**

$$\int e^x \{ (x^3 + 3x^2) + (x^2 + 2x) + (x + 1) \} dx$$

$$= e^x (x^3 + x^2 + x) + C$$

$$\text{Hence } f(2) = 8 + 4 + 2 = 14$$

7. Ans (3.00)



$R \rightarrow$ Radius of sphere

$r \rightarrow$ Radius of cylinder

$h \rightarrow$ height of cylinder

$$R^2 = r^2 + \frac{h^2}{4}$$

Volume of cylinder $= \pi r^2 h$

$$= \pi h \left(R^2 - \frac{h^2}{4} \right) = f(h) \text{ (say)}$$

So

$$f'(h) = \pi \left(R^2 - \frac{3h^2}{4} \right) = 0 \Rightarrow h^2 = \frac{4R^2}{3}$$

$$f''(h) = -3\pi h < 0 \quad \forall h > 0$$

$$\text{So } V_1 = \pi \cdot \frac{2R^2}{3} \cdot \frac{2}{\sqrt{3}} \cdot R = \frac{4}{3\sqrt{3}} \pi R^3$$

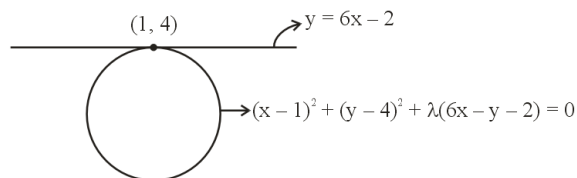
$$V_2 = \frac{4}{3} \pi R^3$$

$$\text{So } \frac{V_2}{V_1} = \sqrt{3}$$

8. Ans (24.50)

Equation of tangent to $y = x^3 + x^2 + x + 1$ at $(1, 4)$ is

$$y - 4 = 6(x - 1) \Rightarrow y = 6x - 2$$



It passes through origin, so

$$17 - 2\lambda = 0 \Rightarrow \lambda = \frac{17}{2}$$

Hence equation is

$$2x^2 - 4x + 2 + 2y^2 - 16y + 32 + 102x - 17y - 34 = 0$$

$$\Rightarrow 2x^2 + 2y^2 + 98x - 33y = 0$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

01-08-2021

JEE(Main + Advanced) : Enthusiast Course (PHASE : TIAS, S,I(A),I, LIVE-I & I(A))

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,D	A,B	B,C	B,C	A,C	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	D	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	10.16 to 10.17	11.00	2.60	3.75	36.00	3.20	9.50	31.25

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C	A,C,D	A,C,D	A,C	A,C	B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	B	A	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	0.00	0.00	9.00	- 3.22	2.50	5.00	4.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,C	B,C	A,D	A,C	A,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	B	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	49.00	3.00	6.18 or 6.19	3.33 or 3.34	1.00	1.00	7.50	1.04 or 1.05

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

2. Ans (A,B)

$$P^2V = \text{constant} [PV = nRT, P \propto T/V]$$

$$\Rightarrow \frac{T^2}{V^2} V = C = T_V^{2-1}$$

$$\Rightarrow T_0^2 V_0^{-1} = (3V_0)^{-1} T^2$$

$$3T_0^2 = T^2$$

$$\sqrt{3}T_0 = T$$

$$\rightarrow dU = nC_v dT [dT = \sqrt{3}T_0 - T_0 = \text{positive}]$$

So, $dU = +ve = \text{internal energy increase.}$

3. Ans (B,C)

There is a decrease in volume during melting on an ice slab at 273K. Therefore, negative work is done by ice water system on the atmosphere or +ve work is done on the ice-water system by the atmosphere.

Heat is absorbed during melting ($dQ = +ve$) and work done by ice water system is -ve ($dw = -ve$)

Therefore, from first law of thermodynamics.

$$dU = dQ - dW = +ve$$

Change in internal energy of ice water system is +ve.

So, internal energy will increase.

6. **Ans (A,B)**

If $n_1 > n_2$, rays will refract away from the normal

$$\Rightarrow \text{virtual image } \frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

If $n_2 = 2n_1$, then

$$\frac{2n_1}{v} - \frac{n_1}{u} = \frac{n_1}{R}$$

For image to be virtual $v < 0 \Rightarrow u < R$

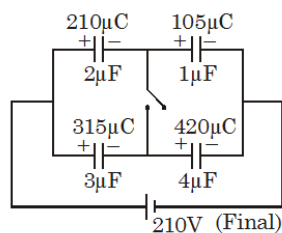
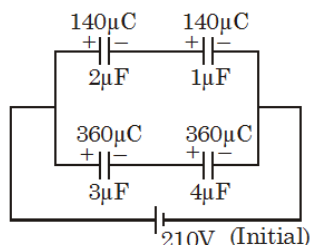
SECTION-I (ii)

8. **Ans (B)**

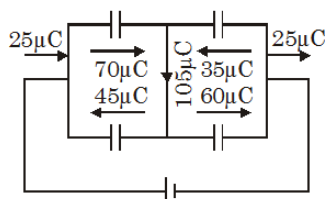
Apply KVL

9. **Ans (D)**

The charges on the capacitors in initial and final conditions can be calculated as shown in the figure.



Comparing the initial and final conditions, the charges flowing through different branches can be obtained as shown in the figure.



Therefore, the charge flowing through the switch is $105 \mu\text{C}$.

Increase in energy stored in capacitor

$$\Delta u_C = (\text{Final energy stored}) - (\text{Initial energy stored})$$

$$= \left[\frac{1}{2} \left(\frac{5}{2} \right) (210)^2 - \frac{1}{2} \left(\frac{50}{21} \right) (210)^2 \right] \mu\text{J}$$

Extra energy supplied by battery

$$\Delta u_B = 25 \times 210 \mu\text{J} = 5.25 \text{ mJ}$$

$$\Delta H = \Delta u_B - \Delta u_C$$

$$\text{Heat loss,} = (5.25 - 2.625) \text{ mJ} = 2.625 \text{ mJ}$$

10. **Ans (A)**

$$\text{We have } V_A = \frac{1}{4\pi\epsilon_0} \left(\frac{q_A}{r_A} - \frac{q_B}{r_B} \right) \dots (1)$$

$$V_B = \frac{1}{4\pi\epsilon_0} \left(\frac{q_A}{r_B} - \frac{q_B}{r_B} \right) \dots (2)$$

$$V_A - V_B = \frac{q_A}{4\pi\epsilon_0} \left(\frac{1}{r_A} - \frac{1}{r_B} \right) \dots (3)$$

From equation (3), potential difference

depends on q_A only. Further, if q_A is

positive, then $V_A - V_B$ is positive as $r_A < r_B$.

Therefore $V_A > V_B$ So if the two spheres are

connected by a conducting wire, (charge

flows from higher potential to lower

potential) till $V_A = V_B$ or $V_A - V_B = 0$.

Hence, potential difference will become zero

only when $q_A = 0$, i.e., all charge flows from

inner sphere to outer sphere.

If q_A is negative, $(V_A - V_B)$ is negative, i.e.,

$V_A < V_B$. Hence, when the two spheres are

connected by a thin wire all charge q_A will

flow from inner sphere to the outer sphere,

since negative charge flows from lower

potential to higher potential. Thus we see

that the whole charge q_A flows from inner

sphere to outer sphere, no matter how high

q_B is.

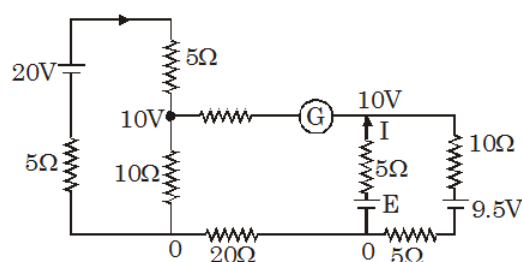
Thus charge flows from A to B, whether q_A

$> q_B$ or $q_B > q_A$ or $V_A > V_B$ or $V_A < V_B$.

SECTION-II

1. **Ans (10.16 to 10.17)**

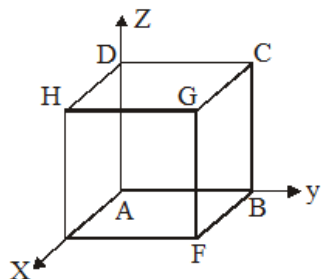
$$\frac{E - 10}{20} = I$$



$$10 = E - \frac{E - 9.5}{20} \times 5$$

$$10 = \frac{3E}{4} + \frac{9.5}{4} \Rightarrow \frac{3E}{4} = 10 - \frac{9.5}{4} = \frac{30.5}{4}$$

2. Ans (11.00)



Flux through ABCD

$$\begin{aligned}\phi_1 &= \vec{E} \cdot \vec{A} \\ &= (x^2\hat{i} + y\hat{j}) \cdot (-a^2\hat{i}) \\ &= 0 \text{ as } x = 0\end{aligned}$$

Flux through EFGH

$$\begin{aligned}\phi_2 &= (x^2\hat{i} + y\hat{j}) \cdot (+a^2\hat{i}) \\ &= x^2, a^2 = a^4 = 1.0 \times 10^{-4} \text{ Nm}^2/\text{C}\end{aligned}$$

Flux through BCGF

$$\begin{aligned}\phi_3 &= (x^2\hat{i} + y\hat{j}) \cdot (a^2\hat{j}) \\ &= a^3 = 1.0 \times 10^{-3} \text{ Nm}^2/\text{C}\end{aligned}$$

Flux through EADH

$$\phi_4 = (x^2\hat{i} + y\hat{j}) \cdot (-a^2\hat{j}) = 0 \text{ as } y = 0$$

Flux through ABFE

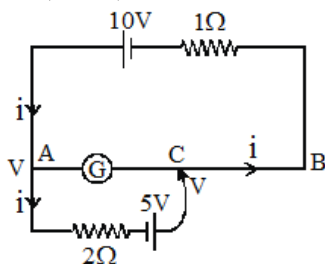
$$\phi_5 = (x^2\hat{i} + y\hat{j}) \cdot (-a^2\hat{k}) = 0$$

Flux through CDHG

$$\phi_6 = 0$$

$$\begin{aligned}\text{Net flux} &= (1.0 \times 10^{-4} + 1.0 \times 10^{-3}) \text{ N-m}^2/\text{C} \\ &= 11 \times 10^{-4} \text{ N-m}^2/\text{C}\end{aligned}$$

3. Ans (2.60)



$$V_A = V_B \therefore 5 - 2i = 0$$

$$i = \frac{5}{2} \text{ A}$$

Now apply KVL

$$9 + 5 = i(1 + 2 + r)$$

$$14 = 2.5(3 + r)$$

$$r = 3\Omega$$

(r is the resistance of BC)

4. Ans (3.75)

The capacitance of a spherical shell of radius R is given by $C = 4\pi\epsilon_0 R$

The charge on the shell is $Q = (4\pi\epsilon_0 R) V$

$$(\because Q = CV) \dots (i)$$

Let R' be the radius of each of the two drops.

The volume of each drop is $\frac{4}{3}\pi(R')^3$. Total

$$\text{volume of the two drops} = 2 \times \frac{4}{3}\pi(R')^3$$

$$\text{Mass of the two drops} = 2 \times \frac{4}{3}\pi(R')^3 \times \rho,$$

where ρ is the density of the liquid inside. By law of conservation of mass,

$$4\pi R^2 t \rho = 2 \times \frac{4}{3}\pi(R')^3 \rho$$

$$\therefore R' = \left(\frac{3R^2 t}{2} \right)^{\frac{1}{3}}$$

Since the liquid is conducting, all of the charge will reside on the outer sphere of each drop. The capacitance of each drop is given by $4\pi\epsilon_0 R'$. Let V' be the potential of each drop. Then charge on the two drops together is $2 \times (4\pi\epsilon_0 R') V'$

By charge conservation,

$$(4\pi\epsilon_0 R)V = 2 \times (4\pi\epsilon_0 R')V'$$

$$\begin{aligned}\therefore V' &= \frac{RV}{2R'} \\ &= \frac{RV}{2 \times \left(\frac{3R^2 t}{2} \right)^{\frac{1}{3}}}\end{aligned}$$

PART-2 : CHEMISTRY

SECTION-I (i)

5. Ans (A,C)

$$\begin{aligned}\Delta H &= 3 \times (-390) - 2(-820) \\ &= 470 \text{ kJ/mol}\end{aligned}$$

$$\Delta G < 0 \quad \Delta H - T\Delta S < 0$$

for spontaneous reaction $\rightarrow T\Delta S > \Delta H$

$$(\because \Delta S > 0)$$

SECTION-II

2. Ans (0.00)

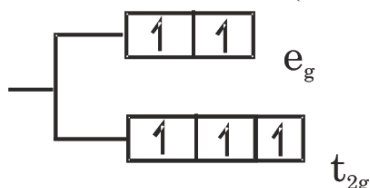
PbS $\longrightarrow$ PbO (on roasting). There is no difference in O.S.

3. Ans (0.00)

$$\mu = \sqrt{35}$$

$\Rightarrow n = 5$ (no. of unpaired e^-)

Fe^{3+} in octahedral field (for $n = 5$)



$$\begin{aligned} \text{C.F.S.E.} &= -0.4\Delta_o n_{t_{2g}} + 0.6\Delta_o n_{e_g} + mp \\ &= -0.4\Delta_o \times 3 + 0.6\Delta_o \times 2 + 0 \\ &= 0 \end{aligned}$$

5. Ans (-3.22)

$$\Delta_r G = RT \ln$$

$$\begin{aligned} \frac{p_{H_2O}}{V \cdot p_{H_2O}} &= RT \ln(0.01) \\ &= \frac{2 \times 350}{1000} \times 2.303(-2) \\ &= -3.22 \text{ kcal/mole} \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I (i)

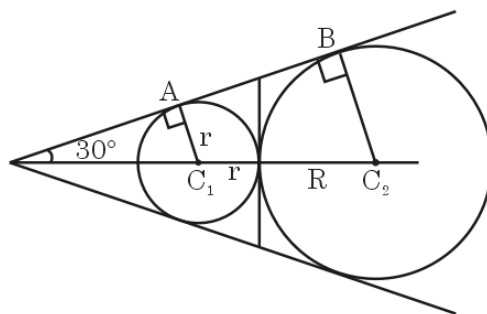
1. Ans (A,C)

$$\begin{aligned} \int e^{-y} dy &= \int x e^{-x^2} dx \\ \Rightarrow -e^{-y} &= -\frac{e^{-x^2}}{2} + C \\ x=0 \text{ \& } y=0 &\Rightarrow C = -\frac{1}{2} \\ \Rightarrow e^{-y} &= \frac{1+e^{-x^2}}{2} \\ \Rightarrow y \text{ or } F(x) &= -\ln\left(\frac{1+e^{-x^2}}{2}\right) \\ \Rightarrow F'(x) &= \frac{+2 \cdot e^{-x^2} \cdot 2x}{(1+e^{-x^2})} \\ &\quad \text{--- -- -- + +} \\ &\quad \text{O} \\ &\quad \text{min} \end{aligned}$$

2. Ans (A,C)

$$\begin{aligned} \int \cos x \cos x dx &= \int \cos^2 x dx \\ &= \int \left(\frac{1+\cos 2x}{2}\right) dx \\ &= \frac{1}{2}x + \frac{\sin 2x}{4} + C \end{aligned}$$

3. Ans (B,C)



$$\sin 30^\circ = \frac{R-r}{R+r}$$

$$\Rightarrow R = 3r$$

$$PC_1 = 2r \Rightarrow h = 3r$$

$$\text{Area of } \Delta = \frac{h^2}{\sqrt{3}} = \frac{9}{\sqrt{3}} r^2 = 9\sqrt{3}$$

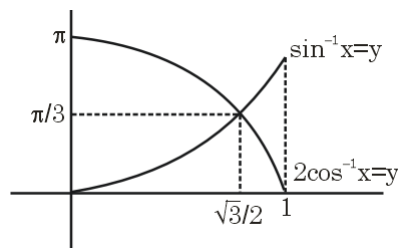
$$\Rightarrow r = \sqrt{3}$$

$$\begin{aligned} L_{DCT} &= \sqrt{(R+r)^2 - (R-r)^2} \\ &= \sqrt{4Rr} = 2\sqrt{3}r = 6 \end{aligned}$$

4. Ans (A,D)

$$\begin{aligned} I_1 &= \int_0^1 e^x \left(x^2 - \frac{x}{F} + 2 + 2x \cdot \frac{1}{F}\right) dx \\ &= e^x \left(x^2 - x + 2\right) \Big|_0^1 = 2e - 2 \\ \& \ I_1 + I_2 &= 3e \cdot 1 - 1 \cdot 0 = 3e \\ \Rightarrow I_2 &= e + 2 \end{aligned}$$

5. Ans (A,C)



$$A_1 + A_2 = \int_0^{\pi/2} \cos \frac{y}{2} dy = 2$$

$$A_2 = \int_0^{\pi/3} \sin y dy + \int_{\pi/3}^{\pi/2} \cos \frac{y}{2} dy$$

$$= \frac{1}{2} + 1 = \frac{3}{2}$$

$$A_1 = \frac{1}{2}$$

6. Ans (A, D)

$$y = e^{x^{e^{e^{e^{e^{e^{\dots}}}}}}}}$$

$$\Rightarrow \log y = x^{e^y} \Rightarrow \log \log y = x^{e^y} \cdot \log x$$

$$\Rightarrow \log \log y = e^{(e^y \log x)} \cdot (\log x)$$

Diff. both sides

$$\frac{1}{\log y} \cdot \frac{1}{y} \cdot y' = e^{(e^y \log x)} \left\{ \frac{e^y}{x} + \log x \cdot e^y \cdot y' \right\} \cdot \log x + e^{(e^y \log x)} \cdot \frac{1}{x}$$

$$\text{at } x = 1 \quad y = e$$

$$\frac{1}{1} \cdot \frac{1}{e} \cdot y' = 1 \left\{ \frac{e^e}{1} + 0 \right\} \cdot 0 + \frac{1}{1}$$

$$\therefore y' = e$$

SECTION-I (ii)

7. Ans (A)

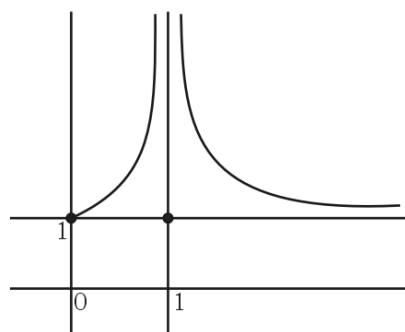
$$F(x) = \frac{(x+1)^2}{(x^2+1)} \times \frac{(x^2+1)^2}{(x^4+1)} \times \frac{(x^4+1)^2}{(x^8+1)} \times \dots$$

$$F(1) = F(0) = 1$$

$$F(x) = \lim_{n \rightarrow \infty} \frac{(x+1)^2 (x^2+1) (x^4+1) \dots (x^{2^{n-1}}+1)}{(x^{2^n}+1)}$$

$$F(x) = \lim_{n \rightarrow \infty} \frac{(x+1)^2}{(x^2-1)} \cdot \frac{(x^{2^n}-1)}{(x^{2^n}+1)}$$

$$= \begin{cases} 1 & x = 0 \\ \frac{-(x+1)}{(x-1)} & 0 < x < 1 \\ 1 & x = 1 \\ \frac{x+1}{x-1} & x > 1 \end{cases}$$



8. Ans (D)

$$(P) \int_1^5 \sqrt{x-2} \sqrt{x-1} \, dx = \int_1^5 |\sqrt{x-1} - 1| \, dx$$

$$= \int_1^2 (1 - \sqrt{x-1}) \, dx + \int_2^5 (\sqrt{x-1} - 1) \, dx$$

$$= \left[x - \frac{2(x-1)^{3/2}}{3} \right]_1^2 + \left[\frac{2(x-1)^{3/2}}{3} - x \right]_2^5$$

$$= \left(2 - \frac{2}{3} - 1 + 0 \right) + \left(\frac{16}{3} - 5 - \frac{2}{3} + 2 \right) = -2 + \frac{12}{3} = 2$$

(Q)

$$\int_{-1}^2 \left[\frac{[x]}{1+x^2} \right] dx = \int_{-1}^0 \left[\frac{-1}{1+x^2} \right] dx + \int_0^1 [0] dx + \int_1^2 \left[\frac{1}{1+x^2} \right] dx$$

$$= \int_{-1}^0 (-1) dx + 0 + 0 \Rightarrow -1$$

$$(R) I = \int_{1/2}^2 \frac{x \ln x}{(1+x^2)^2} dx \quad \text{Put } x = \frac{1}{t}$$

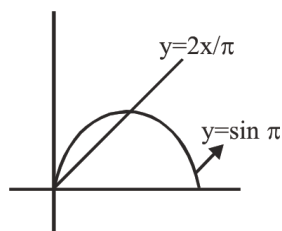
$$\Rightarrow dx = -\frac{1}{t^2} dt$$

$$I = - \int_2^{1/2} \frac{1}{t} \frac{\ln t \, t^4}{(t^2+1)^2} \left(-\frac{1}{t^2} \right) dt$$

$$\Rightarrow I = -I$$

$$\Rightarrow I = 0$$

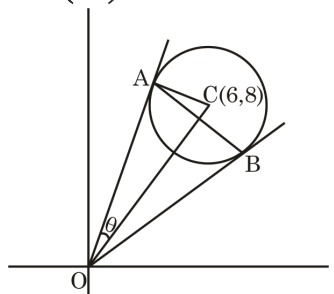
$$(S) \int_0^{\pi} \left| \sin x - \frac{2x}{\pi} \right| dx$$



$$\Rightarrow \int_0^{\pi/2} \left(\sin x - \frac{2x}{\pi} \right) dx + \int_{\pi/2}^{\pi} \left(\frac{2x}{\pi} - \sin x \right) dx$$

$$= \left[-\cos x - \frac{x^2}{\pi} \right]_0^{\pi/2} + \left[\frac{x^2}{\pi} + \cos x \right]_{\pi/2}^{\pi}$$

9. Ans (B)



$$OA = \sqrt{c}, CA = \sqrt{100-c}$$

$$\sin \theta = \frac{\sqrt{100-c}}{10}$$

$$AB = 2\sqrt{c} \sin \theta$$

$$5\sqrt{3} = 2\sqrt{c} \cdot \frac{\sqrt{100-c}}{10}$$

$$\Rightarrow c = 75 \text{ or } 25$$

$$\therefore \angle CAO = 90^\circ$$

$$\therefore \text{Circle } S_2 = x^2 + y^2 = 75$$

10. Ans (C)

$$(P) \lim_{x \rightarrow 0} \frac{e^{x^2} + ax + b}{x^2} = \frac{(1 + \frac{x^2}{1!} + \frac{x^4}{2!} + \dots) + ax + b}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{(1+b) + ax + \frac{x^2}{1}}{x^2}$$

$$a + b = 1$$

$$(Q) \text{ For continuity } f(-2) = \lim_{x \rightarrow -2} f(-2+h)$$

$$4p - 2q = -2p + 2$$

$$3p = q + 1$$

$$\text{LHD} = -4p + q$$

$$\text{RHD} = p$$

$$-4p + q = p$$

$$q = 5p$$

$$3p = 5p + 1 \Rightarrow p = -\frac{1}{2} \text{ \& } q = -\frac{5}{2}$$

$$\left| \frac{q}{p} \right| = 5$$

$$(R) \lim_{x \rightarrow 0} \sqrt{\frac{1 - \cos \frac{x}{t}}{\left(\frac{x^2}{t^2}\right)t^2}} = \lim_{x \rightarrow 0} \frac{\cos\left(\frac{\pi}{2} + h\right)}{-h} \cdot \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{1}{\sqrt{2}|t|} = \frac{1}{\sqrt{2}} \Rightarrow t = \pm 1$$

$$(S) \text{ Area of DABC} = \Delta = \frac{1}{2} r^2 \sin 2r$$

$$V = \frac{4}{3} \pi r^3$$

$$\lim_{r \rightarrow 0} \frac{16\pi\Delta}{3V} = \lim_{r \rightarrow 0} \frac{\frac{1}{2} r^2 \sin 2r}{\frac{4}{3} \pi r^3} \left(\frac{16\pi}{3} \right) = 4$$

SECTION-II

1. Ans (49.00)

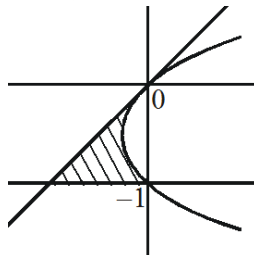
$$f'(x) = (x^2-1)(x^2-2)(x^2-4)^3 2x$$

$$= 2x(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})(x-2)^3(x+2)^3$$

$$\therefore a = 7$$

$$\Rightarrow a^2 = 49$$

2. Ans (3.00)



$y^2 + y = x$ is plotted as in adjacent diagram

$$2yy' + y' = 1$$

$$y' = \frac{1}{2y+1}$$

$$\frac{dy}{dx} \Big|_{(0,0)} = 1$$

equation of tangent at origin is $y = x$

Now required area (Horizontal strip) is

$$\Delta = \int_{-1}^0 [(y^2 + y) - y] dy = \frac{1}{3}$$

3. Ans (6.18 or 6.19)

$$I = \int \left(\frac{\sin x}{\cos x} \right)^{-3/4} (\cos x)^6 dx$$

$$= \int (\tan x)^{-3/4} (\sec)^6 dx$$

$$\text{put } \tan x = t$$

$$I = \int t^{-3/4} (1+t^2)^2 dt$$

$$= \int (t^{-3/4} + 2t^{5/4} + t^{13/4}) dt$$

$$= \frac{t^{1/4}}{1/4} + \frac{2t^{9/4}}{9/4} + \frac{t^{17/4}}{17/4} + c$$

$$a_1 = \frac{(\tan x)^{1/4}}{1/4} + \frac{(\tan x)^{9/4}}{9/8} + \frac{(\tan x)^{17/4}}{17/4} + c$$

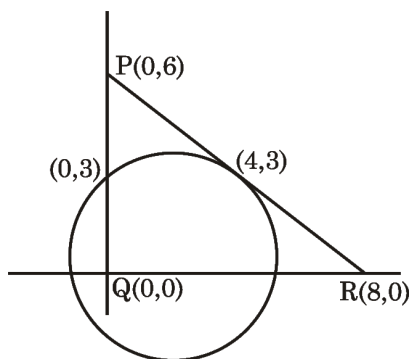
$$a_1 = 1/4 \quad b_1 = 1/4$$

$$a_2 = 9/4 \quad b_2 = 9/8$$

$$a_3 = 17/4 \quad b_3 = 17/4$$

$$a_1 + a_2 + a_3 + b_1 + b_2 + b_3 = \frac{99}{8}$$

4. Ans (3.33 or 3.34)



Circle is $(x - 4)^2 + (y - 3)^2 + \lambda \left(\frac{x}{8} + \frac{y}{6} - 1 \right) = 0$

passes through (0, 3)

$$\Rightarrow 16 + \lambda \left(-\frac{1}{2} \right) = 0 \Rightarrow \lambda = 32$$

$$\Rightarrow x^2 + y^2 - 4x - \frac{2}{3}y - 7 = 0$$

$$r = \sqrt{4 + \frac{1}{9} + 7} = \frac{10}{3}$$

5. Ans (1.00)

$$I_1 = \int_{\pi/6}^{\pi/3} \tan^2 \theta f(2 \operatorname{cosec} 2\theta) d\theta \quad (\text{Apply King})$$

$$= \int_{\pi/6}^{\pi/3} \cot^2 \theta f(2 \operatorname{cosec} 2\theta) d\theta = I_2 \Rightarrow \frac{I_1}{I_2} = 1$$

6. Ans (1.00)

$$2 \tan^{-1} (\sqrt{10} - 3) = \tan^{-1} \left(\frac{2(\sqrt{10} - 3)}{1 - (\sqrt{10} - 3)^2} \right)$$

$$= \tan^{-1} \left(\frac{2(\sqrt{10} - 3)}{6\sqrt{10} - 18} \right)$$

$$= \tan^{-1} \left(\frac{1}{3} \right) \text{ and}$$

$$\cos^{-1} \left(\frac{2}{\sqrt{5}} \right) = \tan^{-1} \left(\frac{1}{2} \right)$$

$$\Rightarrow \tan \left(\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} \right) = \tan \frac{\pi}{4} = 1$$

7. Ans (7.50)

$$f'(x) = xf(x)$$

$$\Rightarrow y = f(x) = ce^{\frac{x^2}{2}}$$

Let $f(x)$ touches $y = x$ at point (α, α)

$$\Rightarrow f(\alpha) = \alpha \text{ \& } f'(\alpha) = 1$$

$$\Rightarrow \alpha f(\alpha) = 1 \text{ or } \alpha = 1 \quad (\because f(x) > 0)$$

$$f(1) = 1 \Rightarrow c = \frac{1}{\sqrt{e}}$$

$$f(4) = \frac{1}{\sqrt{e}} e^8 \Rightarrow \ln f(4) = \frac{15}{2} = 7.50$$

8. Ans (1.04 or 1.05)

$$\ell = \lim_{n \rightarrow \infty} \sum_{r=1}^{n-1} \sqrt{\frac{r}{n^3 - r^3}}$$

$$\ell = \lim_{n \rightarrow \infty} \sum_{r=1}^{n-1} \frac{1}{n} \sqrt{\frac{\frac{r}{n}}{1 - \left(\frac{r}{n}\right)^3}}$$

$$\ell = \int_0^1 \frac{\sqrt{x}}{\sqrt{1 - x^3}} dx$$

$$\text{put } x^{\frac{3}{2}} = t \Rightarrow \frac{3}{2} \sqrt{x} dx = dt$$

$$\ell = \frac{2}{3} \sin^{-1} t \Big|_0^1 = \frac{\pi}{3} \approx 1.047$$